STUDYLAB HUB · AP & COLLEGE GENERAL CHEMISTRY · UNIT 2 (CHAPTERS 3–4)

Unit 2: Chemical Bonding & Molecular Geometry

Ionic vs. Covalent · Lattice Energy · Formal Charge · Resonance · VSEPR 3D Shapes & sp/sp²/sp³ Hybridization

AP & College General Chemistry — Unit 2 (Chapters 3–4): Chemical Bonding, Lewis Structures, VSEPR & Hybridization

Independent, original study guide for AP & College General Chemistry (informed by OpenStax open textbooks, CC BY 4.0).

1. Core Unit Roadmap

Section	Core Principle & Definitions	High-Yield Exam Application
Part 1 · Fundamental Concepts	Ionic vs. Covalent	Identifying core variables, boundary conditions, and conserved quantities
Part 2 · Quantitative Models & Mechanisms	Lattice Energy	Applying governing equations and predicting how system state changes under perturbation
Part 3 · Advanced Problem Solving	VSEPR 3D Shapes & sp/sp^2/sp^3 Hybridization	Avoiding common sign, unit, and conceptual traps on multiple-choice and free-response questions

2. Essential Principles & Step-by-Step Analysis

1. **First-Principles Mechanism**:

- Every problem in **Unit 2 (Chapters 3–4): Chemical Bonding, Lewis Structures, VSEPR & Hybridization** reduces to identifying the underlying structure, forces, rates, or invariants before plugging in numbers.
- Always state the governing principle explicitly at the start of any free-response justification.

2. **High-Yield Comparison & Edge Cases**:

- Distinguish between equilibrium/steady-state behavior and transient/limiting behavior.
- Verify dimensional consistency and limiting cases (e.g., as distance or concentration approaches zero or infinity).

3. **Worked Exam Strategy**:

- Write out the known quantities with units, select the exact relation connecting them, solve symbolically first, and then substitute numerical values.